AC CONTROLLER - 6

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→ **Source Code:**

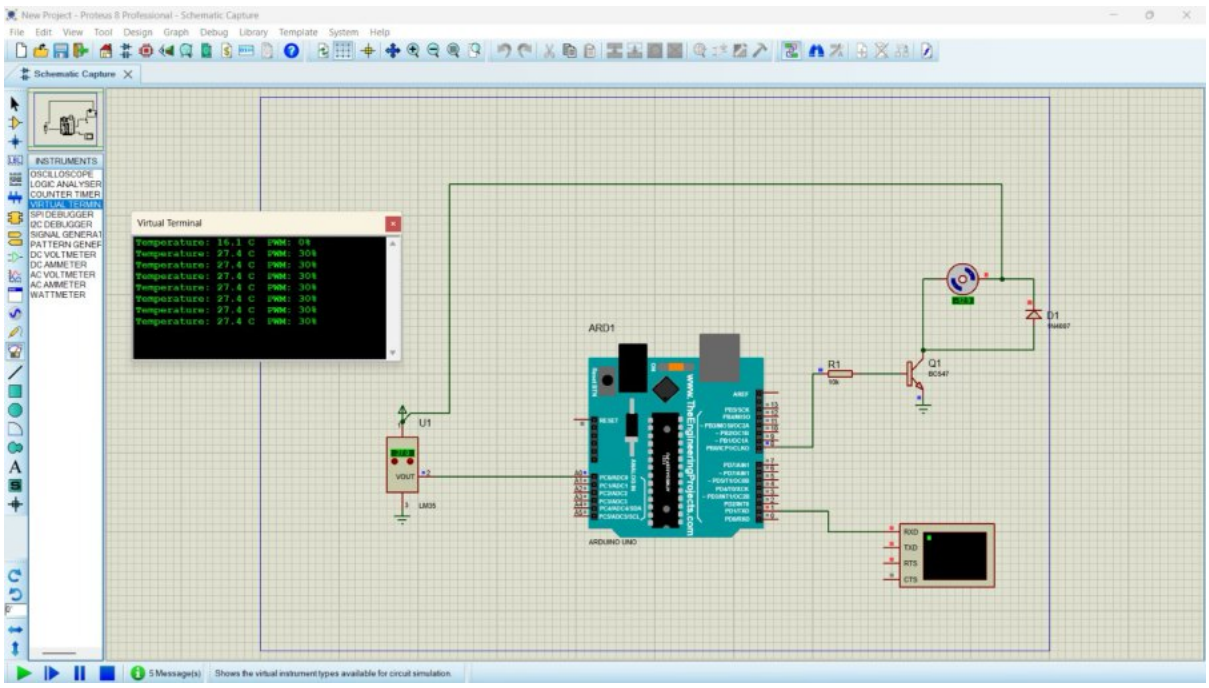
```
#define TEMP A0
#define FAN 8
volatile byte pwm = 0;
volatile byte count = 0;
// Timer1 Compare Match Interrupt
ISR(TIMER1_COMPA_vect)
{
  count++;
  if (count >= 100)
  {
    count = 0;
  }
  digitalWrite(FAN, count < pwm);
}
void setup()
{
  Serial.begin(9600);
```

```
  pinMode(FAN, OUTPUT);
  // Timer1 configuration
  TCCR1A = 0;
  TCCR1B = (1 << WGM12) | (1 << CS11);
  OCR1A = 199;
  TIMSK1 = (1 << OCIE1A);
  sei();
}

void loop()
{
  int adc = analogRead(TEMP);

  // Convert ADC value into temperature (LM35 sensor)
  float temp = adc * 500.0 / 1023.0;
  if (temp < 25)
  {
    pwm = 0;
  }
  else if (temp < 30)
  {
    pwm = 30;
  }
```

```
  else if (temp < 35)
  {
    pwm = 60;
  }
  else
  {
    pwm = 90;
  }
  Serial.print("Temperature: ");
  Serial.print(temp, 1);
  Serial.print(" C PWM: ");
  Serial.print(pwm);
  Serial.println("%");
  delay(1000);
}
```



Reflections:

1. Learned how to interface the LM35 temperature sensor with Arduino Uno to measure temperature.

2. Understood ADC conversion and the process of converting analog sensor readings into temperature values.

3. Gained practical knowledge of Timer1 configuration and Compare Match Interrupts.

4. Learned to generate software-based PWM signals to control fan speed according to temperature variations.

5. Understood how embedded systems use real-time sensor data for automatic temperature monitoring and fan control.